



PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Requirements Specification

Task Manager Application

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Team 4	02-10-2026	Initial SRS preparation	1.0

Introduction

Purpose

The purpose of this Software Requirements Specification is to define the requirements of the Task Manager Application. The system provides a platform for organizing and tracking work progress through task creation, task assignment, task updates, and task monitoring.

Intended Audience

This document is intended for the project team, developers, testers, project evaluators, and users of the Task Manager Application. It describes the system requirements, interfaces, functional requirements, non-functional requirements, and design constraints.

Product Scope

The Task Manager Application is a task management tool used to organize and track work progress. The system allows users to create, view, update, assign, prioritize, and track tasks.

References

1. Software Engineering Mini-Project Deliverables Part-1, 2026.
2. Mini-Project Problem Statement, Section F, Team 4.
3. Project Guidelines, Software Engineering Mini-Project.

Overall Description

Product Perspective

The Task Manager Application is a web-based software system for organizing and tracking work progress. The system provides a user interface for task management and application services for processing and storing user and task information.

Product Functions

The major functions of the product are:

- User registration and authentication.
- User login and logout.
- Create, view, edit, and delete tasks.
- Assign tasks to team members.
- Set task priority and deadline.
- Update and track task status.
- Search and filter tasks.
- View task progress through a dashboard.

User Classes and Characteristics

*Registered User: Creates an account, logs in, manages authorized tasks, and tracks progress.
Team Member: Views assigned tasks, updates task status, and tracks assigned work.*

Operating Environment

The system will operate as a web-based application accessed through a modern web browser. The frontend will be implemented using React.js and the backend will be implemented using Node.js. A database will be used for persistent storage.

Design and Implementation Constraints

The system shall use JavaScript with React.js for the frontend and Node.js for the backend. The system shall use persistent storage for user and task information. Authentication and authorization shall be implemented for protected operations. The system shall follow modular software design principles.

2.6 Assumptions and Dependencies

Users are assumed to have access to a compatible web browser and network connectivity. Users are assumed to provide valid information. The system depends on React.js, Node.js, the selected database system, and the required software libraries and packages.

External Interface Requirements

User Interfaces

The system shall provide a web-based graphical user interface with screens for registration, login, task management, task details, task tracking, and dashboard information. Forms shall validate required input and display appropriate error messages.

Software Interfaces

The React.js frontend shall communicate with the Node.js backend through application APIs. The Node.js backend shall communicate with the database for storing and retrieving user and task information.

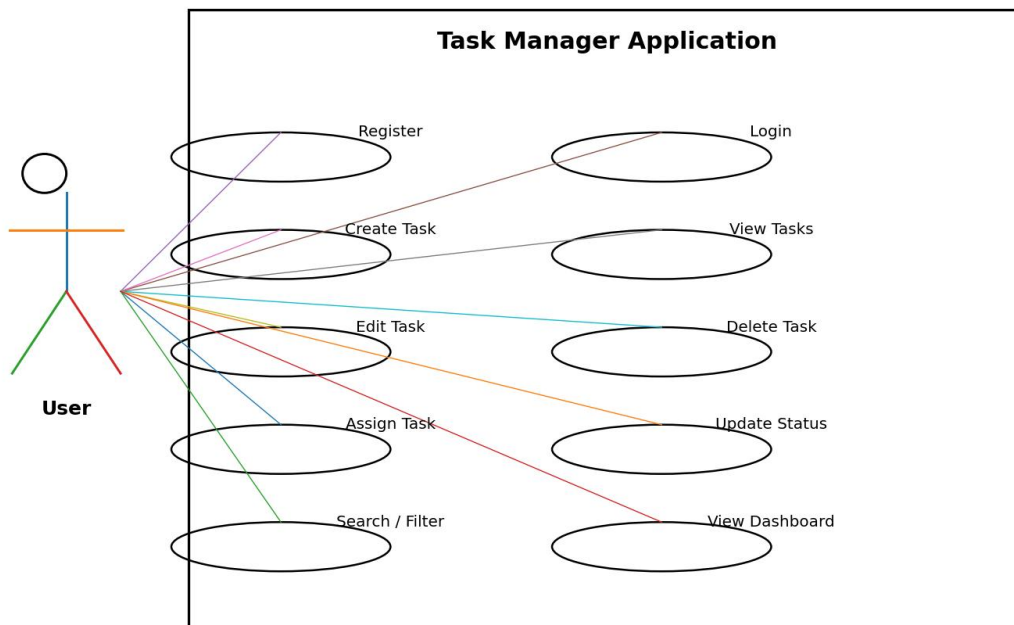
Communications Interfaces

The application shall use HTTP/HTTPS-based client-server communication. API requests and responses shall use JSON or another structured data format. HTTPS shall be used in deployment to protect data transmitted between the client and server.

Hardware Interfaces: No dedicated hardware interface is required.

Analysis Models

The UML use case diagram below identifies the main interactions between the user and the Task Manager Application.



System Features

The functional requirements are organized by the major services provided by the Task Manager Application.

System Feature 1

Authentication and User Management

5.1.1 Description and Priority

Provides user registration, login, logout, and authentication. Priority: High.

5.1.2 Stimulus/Response Sequences

User submits registration details → system validates the details and creates an account. User submits login credentials → system validates the credentials and grants or denies access.

5.1.3 Functional Requirements

*FR-01: The system shall allow a new user to register using a unique email address and password.
FR-02: The system shall authenticate registered users before allowing access to protected functionality.*

Each functional requirement is uniquely identified using a requirement ID.

FR-01: The system shall allow a new user to register using a unique email address and password.

FR-02: The system shall authenticate registered users before allowing access to protected functionality.

System	Feature	2	—	Task	Management
Priority:					High.
FR-03: The system shall allow an authenticated user to create a task with a title, description, priority, and deadline.					
FR-04: The system shall allow an authenticated user to view tasks that the user is authorized to access.					
FR-05: The system shall allow an authorized user to edit an existing task.					
FR-06: The system shall allow an authorized user to delete an existing task.					
FR-07: The system shall allow an authorized user to assign a task to a registered team member.					
FR-08: The system shall allow an authorized user to update the status of a task.					
FR-09: The system shall allow an authorized user to specify and modify task priority.					
FR-10: The system shall allow users to search and filter authorized tasks using task attributes.					
FR-11: The system shall display task progress information through a dashboard.					

Other Nonfunctional Requirements

Performance Requirements

NFR-01: The system shall return normal task-management requests within 3 seconds under normal operating conditions.

Safety Requirements

The system is not intended for safety-critical operations or control of physical equipment. No dedicated safety interface is required.

Security Requirements

Security

Objectives:

- 1. Ensure that only authenticated users can access protected task-management functionality.*
- 2. Protect user and task information from unauthorized access or modification.*

Security

Requirements:

SR-01: The system shall require authentication before allowing access to protected task-management functions.

SR-02: The system shall enforce authorization checks before allowing a user to access or modify a task.

SR-03: The system shall not expose user passwords in application responses or normal application logs.

SR-04: The system shall validate user-supplied input before processing task-management requests.

Software Quality Attributes

NFR-02: The system shall prevent unauthorized users from accessing protected task-management functions.

NFR-03: The system shall provide a consistent interface for creating, viewing, editing, and tracking tasks.

NFR-04: The system shall preserve successfully stored task information during normal application operation.

NFR-05: The system shall use modular components so that individual modules can be modified without requiring changes to unrelated modules.

NFR-06: The system shall structure major functional components so that they can be tested independently.

Business Rules

1. A user must be authenticated before accessing protected task-management functions.

2. Only authorized users may modify or delete a task.

3. A task must contain a title before it can be stored.

4. A task shall have a valid status value.

5. A task assigned to a user shall reference a registered user.

Domain requirements: task information shall include a title, description, priority, deadline, assignee where applicable, and status.

Other Requirements

The system shall maintain persistent records for users and tasks. The implementation shall separate application logic and data storage and shall support future task-management enhancements.

Appendix A: Glossary

Task: A unit of work that needs to be completed.

Priority: The importance level assigned to a task.

Status: The current state of a task.

Dashboard: A view that summarizes task and progress information.

Authentication: Verification of a user's identity.

Authorization: Determination of permitted operations.

API: Application Programming Interface.

FR: Functional Requirement.

NFR: Non-Functional Requirement.

Appendix B: Field Layouts

Field layouts and properties for the Task Manager Application are provided below.

Task information fields and their properties are listed in the table.

Field	Length	Data Type	Description	Is Mandatory
Task ID	20	Alphanumeric	Unique identifier for a task	Y
Title	100	Title of the task		Y
Description	500	String	Detailed task description	N

Priority	10	String	Priority of the task	Y
Deadline	10	Date	Task deadline	N
Status	20	String	Current task status	Y
Assignee	50	String	Assigned team member	N

Task and progress report fields are listed in the report requirements table.

Task Report	Progress Report
Task ID	Total Tasks
Title	To Do Tasks
Priority	In Progress Tasks
Deadline	Completed Tasks
Status	Overdue Tasks
Assignee	Progress Percentage

Appendix C: Requirement Traceability Matrix

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
1	FR-01	User registration	TBD	TBD	TBD	TBD	TBD
2	FR-02	User authentication	TBD	TBD	TBD	TBD	TBD

3	FR-03	Create task	TBD	TBD	TBD	TBD	TBD
4	FR-04	View tasks	TBD	TBD	TBD	TBD	TBD
5	FR-05	Edit task	TBD	TBD	TBD	TBD	TBD
6	FR-06	Delete task	TBD	TBD	TBD	TBD	TBD
7	FR-07	Assign task	TBD	TBD	TBD	TBD	TBD
8	FR-08	Update task status	TBD	TBD	TBD	TBD	TBD
9	FR-09	Task priority	TBD	TBD	TBD	TBD	TBD
10	FR-10	Search and filter	TBD	TBD	TBD	TBD	TBD
11	FR-11	Dashboard	TBD	TBD	TBD	TBD	TBD
12	NFR-01	Performance	TBD	TBD	TBD	TBD	TBD
13	NFR-02	Security	TBD	TBD	TBD	TBD	TBD
14	NFR-03	Usability	TBD	TBD	TBD	TBD	TBD
15	NFR-04	Reliability	TBD	TBD	TBD	TBD	TBD
16	NFR-05	Maintainability	TBD	TBD	TBD	TBD	TBD
17	NFR-06	Testability	TBD	TBD	TBD	TBD	TBD
18	SR-01	Authentication security requirement	TBD	TBD	TBD	TBD	TBD
19	SR-02	Authorization security requirement	TBD	TBD	TBD	TBD	TBD
20	SR-03	Password protection	TBD	TBD	TBD	TBD	TBD
21	SR-04	Input validation	TBD	TBD	TBD	TBD	TBD